LEVEL UP ACADEMY

Class 10 | GSEB | Science

ANSWER KEY & MARKING SCHEME

Ch 8 Test (GSEB pattern): Heredity

Chapter / Topic: Heredity

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SECTION A

Q1. Tall pea plants with violet flowers were crossed with short pea plants with white flowers. All the progeny had violet flowers, but almost half of… **[1]**

Answer: (b) TtWW. Half the progeny are short, so the parent is Tt. All are violet, so the parent is WW.

Q2. Plants with round seeds (RR) are crossed with plants with wrinkled seeds (rr). The F_1 seeds are: **[1]**

Answer: (a) all round (Rr)

Q3. In human beings, the sex of the child is decided by: **[1]**

Answer: (b) the father. The sperm carries either an X or a Y chromosome.

Q4. How many pairs of chromosomes are there in human body cells? **[1]**

Answer: (c) 23 pairs (46 chromosomes)

Q5. In Mendel's monohybrid cross, the phenotypic ratio in the F_2 generation is: **[1]**

Answer: (d) 3 : 1

Q6. A child has the sex chromosomes XX. The child is: **[1]**

Answer: (d) a girl

Q7. Genes are located on the: **[1]**

Answer: (c) chromosomes (as segments of DNA)

Q8. In the F_2 generation of Mendel's dihybrid cross, what fraction of plants show both recessive traits (wrinkled and green)? **[1]**

Answer: (a) 1/16

Q9. When a pure tall (TT) pea plant is crossed with a pure short (tt) plant, the F_1 plants are: **[1]**

Answer: (a) all tall (Tt)

SECTION B

Q10. What are dominant and recessive traits? Give one example of each from Mendel's experiments. **[2]**

Answer: A dominant trait is expressed even when only one copy of its gene is present (e.g. tallness, T). A recessive trait is expressed only when both copies are of the recessive type (e.g. shortness, tt).

- Dominant with example: 1
- Recessive with example: 1

Q11. Why were all the F_1 plants tall, with no medium-height plants, when Mendel crossed pure tall and pure short pea plants? **[2]**

Answer: Tallness (T) is dominant over shortness (t). The F_1 plants are Tt, and a single copy of T is enough to make the plant tall. Traits do not blend, so no medium-height plants appear.

- Tall is dominant, F_1 is Tt: 1
- No blending of traits: 1

Q12. How is the sex of a child determined in human beings? **[2]**

Answer: The mother has two X chromosomes, so every egg carries an X. The father has one X and one Y, so half the sperm carry X and half carry Y. If an X-carrying sperm fertilises the egg, the child is a girl (XX). If a Y-carrying sperm fertilises it, the child is a boy (XY).

- Mother XX gives X; father XY gives X or Y: 1
- XX → girl, XY → boy: 1

SECTION C

Q13. Mendel crossed pea plants with round, yellow seeds (RRYY) with plants having wrinkled, green seeds (rryy). **[3]**

(i) All round and yellow (RrYy). [1]

(ii) 9 round yellow : 3 round green : 3 wrinkled yellow : 1 wrinkled green [1]

(iii) The two traits (seed shape and seed colour) are inherited independently of each other. [1]

Q14. A pea plant with violet flowers (Vv) is crossed with a plant with white flowers (vv). **[3]**

(i) Vv parent: V and v. vv parent: v only. [1]

(ii) Vv and vv in the ratio 1 : 1. [1]

(iii) 1 violet : 1 white [1]

SECTION D

Q15. A student crossed pea plants with violet flowers (VV) with plants having white flowers (vv). All F_1 plants had violet flowers. When the F_1 plants… **[4]**

(i) Violet [1]

(ii) Vv [1]

(iii) The F_2 genotypes are VV : Vv : vv = 1 : 2 : 1, so out of 400 plants we expect 100 VV, 200 Vv and 100 vv. So about 100 of the 300 violet plants are VV. [2]

- 1 : 2 : 1 ratio: 1
- Answer 100: 1